

## KANDI, Benin

WMO: 653060

Lat: 11.133N

Lon: 2.933E

Elev: 958

StdP: 14.19

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 61.3       | 63.3 | 23.7                          | 18.5 | 83.9 | 26.8 | 21.4 | 82.9 | 22.4                  | 81.1 | 19.2 | 83.3 | 0.9                      | 40   | 0.337 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 19.3                   | 103.7           | 71.3 | 102.2 | 71.4 | 100.6 | 71.6 | 81.1                | 91.0 | 80.4 | 90.5 | 79.8 | 89.9 | 6.9                  | 40   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.8                            | 154.8 | 86.7 | 77.9 | 150.1 | 86.1 | 77.1 | 146.4 | 85.6 | 45.7          | 91.3 | 44.9 | 90.5 | 44.2 | 90.3 | 88.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 16.9          | 14.1 | 11.8 | 56.1                       | 107.3 | 1.7                | 2.3 | 54.9                                               | 108.9 | 53.9       | 110.2 | 53.0       | 111.5 | 51.7       | 113.1 |    |    |
| (5)               |      |      | 46.0                       | 84.5  | 1.3                | 1.8 | 45.0                                               | 85.8  | 44.3       | 86.9  | 43.5       | 87.9  | 42.6       | 89.1  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 82.9   | 77.6  | 83.6  | 89.8  | 90.9  | 87.0  | 83.3  | 80.6  | 79.3  | 80.3  | 82.7  | 81.9  | 78.2 |
|                                                                                  | DBStd       | 5.20   | 3.92  | 4.39  | 3.20  | 3.18  | 3.52  | 3.10  | 2.54  | 2.26  | 2.25  | 2.24  | 2.81  | 3.44 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 12012  | 855   | 941   | 1234  | 1226  | 1148  | 998   | 947   | 908   | 909   | 1015  | 956   | 875  |
|                                                                                  | CDD65       | 6537   | 390   | 521   | 769   | 776   | 683   | 548   | 482   | 443   | 459   | 550   | 506   | 410  |
|                                                                                  | CDH74       | 80280  | 4337  | 6906  | 11708 | 12007 | 9682  | 6667  | 4756  | 3697  | 4298  | 5915  | 5808  | 4499 |
|                                                                                  | CDH80       | 42134  | 2108  | 3985  | 7519  | 7776  | 5512  | 3072  | 1636  | 1029  | 1502  | 2651  | 3109  | 2235 |
| Wind                                                                             | WSAvg       | 4.1    | 4.6   | 4.9   | 5.0   | 5.7   | 5.3   | 4.1   | 3.5   | 3.0   | 2.5   | 2.8   | 3.4   | 4.2  |
| Precipitation                                                                    | PrecAvg     | 39.7   | 0.0   | 0.1   | 0.3   | 1.4   | 4.5   | 6.0   | 8.0   | 10.5  | 7.5   | 1.6   | 0.0   | 0.0  |
|                                                                                  | PrecMax     | 56.5   | 0.3   | 1.4   | 2.3   | 6.2   | 9.3   | 10.1  | 15.4  | 21.6  | 13.3  | 8.1   | 0.6   | 0.3  |
|                                                                                  | PrecMin     | 25.8   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 2.4   | 3.7   | 5.0   | 2.6   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 6.4    | 0.0   | 0.2   | 0.5   | 1.3   | 2.1   | 1.9   | 2.3   | 3.3   | 2.4   | 1.5   | 0.1   | 0.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 98.6  | 103.9 | 105.3 | 106.1 | 102.4 | 97.3  | 92.7  | 90.0  | 92.4  | 96.3  | 98.9  | 98.0 |
|                                                                                  |             | MCWB   | 64.2  | 69.3  | 69.7  | 71.3  | 76.2  | 78.3  | 79.0  | 79.2  | 80.0  | 74.7  | 68.1  | 66.2 |
|                                                                                  | 2%          | DB     | 95.5  | 100.6 | 103.6 | 104.0 | 99.7  | 94.6  | 90.5  | 88.2  | 90.1  | 94.3  | 97.2  | 95.7 |
|                                                                                  |             | MCWB   | 63.3  | 68.5  | 69.8  | 72.6  | 77.3  | 78.3  | 78.5  | 78.5  | 79.5  | 75.9  | 69.0  | 65.0 |
|                                                                                  | 5%          | DB     | 93.1  | 98.4  | 102.2 | 102.3 | 97.7  | 92.8  | 88.9  | 86.8  | 88.5  | 92.5  | 95.6  | 93.5 |
|                                                                                  |             | MCWB   | 62.8  | 66.7  | 69.7  | 73.1  | 77.8  | 78.2  | 78.1  | 78.1  | 79.2  | 76.7  | 68.7  | 63.9 |
|                                                                                  | 10%         | DB     | 90.4  | 96.1  | 100.7 | 100.6 | 95.8  | 91.1  | 87.3  | 85.3  | 87.1  | 90.8  | 93.6  | 90.7 |
|                                                                                  |             | MCWB   | 62.2  | 65.5  | 69.4  | 74.2  | 77.8  | 78.1  | 77.6  | 77.6  | 78.7  | 77.0  | 69.2  | 63.1 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 73.6  | 77.2  | 79.6  | 81.2  | 81.9  | 81.1  | 80.5  | 80.9  | 82.2  | 81.4  | 78.1  | 72.7 |
|                                                                                  |             | MCDB   | 89.8  | 94.7  | 95.9  | 97.0  | 94.0  | 91.5  | 89.6  | 87.8  | 88.6  | 89.9  | 91.4  | 89.8 |
|                                                                                  | 2%          | WB     | 68.6  | 74.5  | 76.8  | 79.8  | 80.8  | 80.1  | 79.4  | 79.5  | 80.9  | 80.4  | 76.4  | 69.3 |
|                                                                                  |             | MCDB   | 86.0  | 92.0  | 94.0  | 96.3  | 93.4  | 90.5  | 88.2  | 86.4  | 87.8  | 89.0  | 88.8  | 89.0 |
|                                                                                  | 5%          | WB     | 64.6  | 71.9  | 75.6  | 78.7  | 79.9  | 79.5  | 78.6  | 78.6  | 80.0  | 79.6  | 74.9  | 66.2 |
|                                                                                  |             | MCDB   | 89.5  | 90.2  | 93.1  | 95.4  | 92.5  | 89.8  | 87.2  | 85.3  | 86.9  | 88.5  | 87.0  | 89.6 |
|                                                                                  | 10%         | WB     | 62.8  | 68.5  | 74.3  | 77.8  | 79.1  | 78.8  | 77.9  | 77.8  | 79.0  | 78.8  | 73.3  | 64.2 |
|                                                                                  |             | MCDB   | 89.0  | 92.6  | 92.4  | 93.9  | 91.7  | 89.0  | 85.9  | 84.1  | 86.0  | 87.9  | 85.1  | 87.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 26.0  | 24.7  | 22.4  | 19.3  | 16.3  | 14.5  | 12.6  | 11.3  | 13.7  | 17.9  | 26.4  | 27.4 |
|                                                                                  |             | MCDBR  | 29.6  | 26.9  | 24.3  | 21.9  | 18.9  | 16.6  | 14.8  | 13.6  | 15.9  | 20.0  | 27.7  | 30.0 |
|                                                                                  |             | MCWBR  | 11.7  | 11.1  | 10.6  | 7.9   | 5.6   | 5.5   | 5.6   | 6.0   | 7.8   | 6.7   | 8.8   | 10.9 |
|                                                                                  | 5% WB       | MCDBR  | 27.3  | 21.4  | 20.2  | 18.5  | 15.9  | 15.1  | 14.0  | 12.8  | 15.3  | 17.0  | 22.5  | 28.2 |
|                                                                                  |             | MCWBR  | 13.3  | 11.3  | 10.2  | 6.4   | 6.2   | 6.6   | 6.1   | 6.4   | 8.4   | 7.5   | 8.8   | 11.9 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.645  | 0.722 | 0.787 | 0.730 | 0.645 | 0.592 | 0.539 | 0.522 | 0.530 | 0.608 | 0.564 | 0.552 |      |
|                                                                                  | taud        | 1.674  | 1.528 | 1.443 | 1.607 | 1.749 | 1.881 | 2.063 | 2.104 | 2.059 | 1.833 | 1.844 | 1.853 |      |
|                                                                                  | Ebn at Noon | 216    | 206   | 196   | 207   | 221   | 230   | 244   | 252   | 252   | 230   | 235   | 236   |      |
|                                                                                  | Edh at Noon | 77     | 92    | 102   | 86    | 73    | 63    | 53    | 52    | 54    | 67    | 65    | 63    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1747   | 1828  | 1896  | 1934  | 1869  | 1761  | 1617  | 1509  | 1680  | 1841  | 1821  | 1736  |      |
|                                                                                  | RadStd      | 71     | 69    | 88    | 65    | 62    | 68    | 58    | 72    | 74    | 64    | 48    | 73    |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page